

Orbit Wars — RL Math, v9: the mixed 2p/4p league with per-seat categorical ownership (*diff* against rl_design_v7.tex)

2026-06-10

How to read this. This document is a *delta* on top of `rl_design_v7.tex` (itself a delta on `rl_math_v6.tex`, itself a delta on `rl_math.tex`). Everything v7 established — BC warm-start, categorical WHERE + Bernoulli gate, gate-only entropy, the three PPO guardrails, potential shaping, PopArt, the GPU simulator — carries over unchanged and is not restated. v9 answers the `it1041` post-mortem: *a checkpoint can dominate heads-up (1.000 gauntlet, 0.82 vs. the previous best) and still lose 4-player free-for-all (29% top-rate vs. 58%), and the Kaggle leaderboard mixes both formats*. The fix is one training program with two coupled parts:

- the **v8 league program** (notebook `setup1_v8_a100.ipynb`): an N -player environment with the official 4p seating and combat, mixed 2p/4p iterations scheduled by a **dual correlated Elo**, invited anchored members, weak-point matchmaking, and a mixed best-checkpoint gauntlet — all *legacy-compatible* (observation unchanged, $F=101$; v7 weights drop in);
- the **v9 observation** (notebook `setup1_v9_a100.ipynb` = v8 + one change): ownership becomes a **4-channel ego-relative seat one-hot** ($F=104$) so the policy can tell *which* opponent owns each body. v8 is kept as the ablation/compatibility lineage; v9 is the design intended to learn per-enemy behavior in FFAs.

One-line map.

component	status in v9	where
simulator dynamics / comets / parity	unchanged 2p; N-player combat added	§1
observation: ownership feature	replaced: seat one-hot ($F\ 101 \rightarrow 104$)	§2
observation: threat block, globals	unchanged (pooled friend/foe)	§2
action space, decode, greedy	unchanged (see <code>rl_design_v7.tex</code> § cat)	—
BC warm-start, PPO guardrails	unchanged (see <code>rl_design_v7.tex</code> § bc, guard)	—
reward: potentials, outcome	generalized to N players	§3
rollout	unified 2p/4p collect_ppo (seats)	§4
league: ratings	dual correlated Elo (2p + 4p)	§5
league: match-type mix	new: gap-driven, error-diffused	§6
league: membership	new: invited members, weak-point boost	§7
best-checkpoint rule	replaced: mixed 2p+4p gauntlet	§8
anchor Elo scale	recalibrated to the leaderboard	§9
deployment adapter	constraint: seat-preserving	§10

1 N -player environment

The board, production, orbits, launches, swept collision, sun occlusion, and official comets are exactly see `rl_math.tex` § “**Batched GPU simulator**” (`sec:gpuenv`) with the v6 data path (see `rl_math_v6.tex` § `sys`). v8/v9 add N -player play ($N \in \{2, 4\}$) without touching the 2p path.

Seating. The official engine assigns homes on one random symmetric group of four planets. `generate_world` records that group’s base index b ; `reset(worlds, n_players)` seats owners exactly like the engine’s branches:

$$N=2: \text{owner}(b) = 0, \text{owner}(b+3) = 1; \quad N=4: \text{owner}(b+j) = j, j = 0..3; \quad S_{\text{home}} = 10.$$

Under the board’s 4-fold rotational symmetry every seat is fair in both formats.

Combat: official top-vs-second, N players. Arrivals at planet e are summed per player, $A_e^{(p)} = \sum_{\text{fleets} \rightarrow e, \text{owner}=p} n$, and the official resolve is the top-two statistic

$$\text{surv}_e = A_e^{(1\text{st})} - A_e^{(2\text{nd})}, \quad \text{owner}_e^{\text{surv}} = \begin{cases} \arg \max_p A_e^{(p)} & \text{surv}_e > 0 \\ \emptyset & \text{tie,} \end{cases} \quad (1)$$

implemented as a (B, E, N) scatter-add + `topk(2)`. For $N=2$ this is *algebraically identical* to the v6 pairwise max/min resolve; survivors reinforce or attack the garrison exactly as before. **Verified:** `scripts/test_v8_parity.py` steps v7 and v8 side-by-side 70 ticks (comet spawn included) with identical actions — planet/fleet state and encodings bit-exact; `scripts/test_v9_parity.py` repeats v8 vs. v9 (state bit-exact; encodings differ only by §2, consistently).

Seat interface. `env_step(env, a, seats=[...])` takes, per opponent pid $1..N-1$, either a scripted code (the v6/v7 `opponent_action`, now seat-parameterized) or a neural gated-alloc action decoded exactly like the ego’s. Termination: step cap, ≤ 1 player alive, or ego death.

2 Observation: per-seat categorical ownership (the v9 change)

`rl_math.tex` § `sec:obs` encodes ownership as one scalar body feature $b_7 \in \{0, 1, 2\}$ (neutral/ego/enemy), computed ego-relatively with $n_a=2$ hard-coded. That scalar cannot represent *which* enemy owns a body, and the formula does not extend: fed a 4-player state, $(o - \text{ego}) \bmod 2$ **aliases seat ego+2 onto the ego code** — the policy would read an opponent’s planets as its own. v8 therefore *collapses* every non-ego owner to the single code 2 (the deployed everyone-vs-me adapter view; $F=101$, v7 weights drop in). v9 replaces the scalar entirely.

Seat one-hot. With $N = \text{n_players}$ and ego seat g , define the ego-relative seat index and the 4-channel categorical

$$\rho_e = (o_e - g) \bmod N, \quad \mathbf{b}_7[e] = \left(\underbrace{\mathbf{1}[o_e \geq 0 \wedge \rho_e = 0]}_{\text{self}}, \mathbf{1}[o_e \geq 0 \wedge \rho_e = 1], \mathbf{1}[o_e \geq 0 \wedge \rho_e = 2], \mathbf{1}[o_e \geq 0 \wedge \rho_e = 3] \right), \quad (2)$$

i.e. $[\text{self} | \text{enemy seat} + 1 | + 2 | + 3]$ with **neutral** = **all-zero**. $N_{\text{body}} 11 \rightarrow 14$, so $F = 101 \rightarrow 104$. Properties that motivated the design:

- **categorical, not ordinal:** the scalar $2..n_a$ code imposes a fake magnitude and ordering across seats; one-hot channels are permutation-symmetric inputs the network can attach independent behavior to (per-enemy threat assessment, target selection);
- **rotation-equivariant:** the same *relative* seat always lands in the same channel, from any ego seat (verified: ego 0 sees owners 1,2,3 as 0100/0010/0001; ego 2 sees itself as 1000 and the others rotated);
- **format-generalizing:** in 2p only the first two channels ever fire, so the 4p input space is a strict superset — and when a seat is eliminated mid-FFA its channel simply stops firing, so 3-player intermediate states remain well-formed without ever being trained explicitly;
- **no aliasing:** ρ is taken mod N (not mod 2), so no opponent can collide with the ego channel.

Everything else in the observation is unchanged: the incoming-fleet threat block keeps the *signed* friend/foe convention (+1 ego, -1 any opponent), and the global aggregates pool *all* opponents (enemy ships = $\sum_{p \neq g}$, enemy production share vs. the pooled field) — seat identity enters only through Eq. (2).

Legacy members: the obs adapter. v9’s league still admits v7/v8-format checkpoints (invited or seeded; their ckpt config carries $F=101$). They are wrapped in **LegacyObsAdapter**, which collapses the one-hot back to the scalar they were trained on before their forward:

$$b_7^{\text{legacy}} = \mathbf{b}_7[0] + 2 \min\left(1, \sum_{k=1}^3 \mathbf{b}_7[k]\right) \in \{0, 1, 2\}, \quad (3)$$

exact on every reachable state (the channels are disjoint indicators). The wrapper is unwrapped on league-state save and re-wrapped on load; it is dormant in v8 (dimensions match). The *learner* cannot be warm-started from v7/v8 weights (F changed): v9 runs its own BC warm-start (see **rl_design_v7.tex § bc**; the BC expert is scripted, so nothing else changes).

3 Reward: N -player potentials and outcome

The reward family is v6 potential shaping + terminal outcome (see **rl_math_v6.tex § reward**), generalized so that $N=2$ reproduces it exactly.

Potentials vs. the strongest opponent. With per-player totals $S^{(p)}$ (planets + fleets), owned production $\Pi^{(p)}$, ship-log $\ell(n) = \ln(1+n)/\ln 1000$:

$$\Phi_{\text{ship}}(s) = \ell(S^{(0)}) - \ell\left(\max_{p \geq 1} S^{(p)}\right), \quad \Phi_{\text{prod}}(s) = \frac{\Pi^{(0)} - \max_{p \geq 1} \Pi^{(p)}}{\max(\Pi_{\text{tot}}, 1)}. \quad (4)$$

The max is the right opponent aggregate for a *win-the-game* potential: in an FFA you are measured against whoever currently leads, not the average of the field. For $N=2$ both reduce to the v6 pair. The shaping form $r_t = c[\gamma\Phi(s_{t+1}) - \Phi(s_t)]$, the coefficients, the σ_R scaling, and the policy-invariance argument are unchanged (see **rl_math_v6.tex § reward**).

Pairwise scores and the 4p outcome. At termination, against each seat k with totals $S^{(k)}$:

$$sc_k = \begin{cases} 1 & S^{(0)} > S^{(k)} \\ \frac{1}{2} & S^{(0)} = S^{(k)} \\ 0 & S^{(0)} < S^{(k)}, \end{cases} \quad \Omega = \begin{cases} \text{sign}(S^{(0)} - \max_{p \geq 1} S^{(p)}) & N=2 \text{ or } \text{OUTCOME_4P}=\text{winner}, \\ 2 \cdot \frac{1}{N-1} \sum_k sc_k - 1 & \text{OUTCOME_4P}=\text{placement (default)}. \end{cases} \quad (5)$$

The placement-linear $\Omega \in \{-1, -\frac{1}{3}, +\frac{1}{3}, +1\}$ maps finishing place affinely (beat-all = +1, lose-to-all = -1): a dense, correctly-ordered learning signal where the official top-score-only rule pays a 4p game like a lottery (2nd of 4 = dead loss). The terminal payment is the v6 win/loss magnitude scaled *linearly* by Ω , $O = \Omega \cdot (\Omega > 0 ? W : L)$, so 2p (where $\Omega \in \{\pm 1, 0\}$) pays exactly as before. The per-seat sc_k are also the dual-Elo update signal (§5); the logged `win_rate` in 4p is the strict “beat *every* opponent” indicator $\mathbf{1}[\min_k sc_k \geq 1]$.

4 Unified rollout

One `collect_ppo(env, net, pool, cursor, rng, seats, n_players)` runs both formats. The learner is always player 0 with the usual on-GPU buffers and GPU GAE (see `rl_math.v6.tex § sys`); each opponent seat is either a scripted anchor (fully on-GPU, zero forwards) or a frozen neural member acting per-step under `no_grad` via its own `env_encode(env, pid)` — the same encode function, ego = that seat (Eq. (2) makes this seat-symmetric by construction). A 4p iteration therefore costs the learner’s rollout + up to three no-grad forwards per step; scripted seats are free. `B_4P = B` keeps the learner’s transition count format-independent. Termination, `done`-correct GAE, the outcome scatter, PopArt, and the PPO update are untouched (see `rl_math.v6.tex § ppo`, see `rl_design.v7.tex § guard`).

5 Dual correlated Elo

Heads-up strength and FFA strength are correlated but *not* identical (the `it1041` failure). Every league member — and the learner — carries two ratings (e^2, e^4). Scripted anchors and invited members are **anchored in both** (their ratings never move); snapshots float.

2p update. After a 2p iteration vs. member m with mean learner score $\bar{s} \in [0, 1]$ over the batch, with expectation $\mathbb{E} = (1 + 10^{(e_m^2 - e_\ell^2)/400})^{-1}$:

$$\Delta = K(\bar{s} - \mathbb{E}), \quad e_\ell^2 += \Delta, \quad e_\ell^4 += \kappa \Delta, \quad (\text{non-anchor } m \text{ mirrored: } e_m^2 -= \Delta, e_m^4 -= \kappa \Delta), \quad (6)$$

with $K = \text{ELO_K} = 32$ and the **cross-format coupling** $\kappa = \text{ELO_COUPLING} = 0.25$.

4p update: pairwise decomposition. A 4p FFA vs. seats $\{m_k\}$ becomes $N-1$ pairwise updates at $K_4/(N-1)$ each (the standard multiplayer decomposition — total rating motion per game matches a 2p game at K_4):

$$\Delta_k = \frac{K_4}{N-1} (sc_k - \mathbb{E}_k^{(4)}), \quad e_\ell^4 += \Delta_k, \quad e_\ell^2 += \kappa \Delta_k \quad \text{for each } k, \quad (7)$$

$K_4 = \text{ELO_K}_4 = 32$, $\mathbb{E}^{(4)}$ the logistic in the e^4 ratings. The coupling encodes the prior that the two skills share most of their variance: a 2p result moves the 4p belief a quarter as much (and vice versa), keeping the ratings correlated without forcing them equal — it is exactly their *gap* that drives the scheduler below.

Recalibration. The 2p anchor back-solve is unchanged (see `rl_math.v6.tex § league`); the 4p analog re-grounds e^4 from the fixed FFA trio: with mean pairwise score s vs. `{starter, greedy, intermediate}` and their mean anchored rating $\bar{a}$, $e^4 = \bar{a} + 400 \log_{10}(s/(1-s))$, s clipped to $[0.02, 0.98]$.

6 Gap-driven match-type mix

The share of 4p iterations is a bounded affine function of the dual-rating gap:

$$s_4 = \text{clip}\left(\text{S4_BASE} + \text{S4_GAIN} \cdot \frac{e_\ell^2 - e_\ell^4}{400}, \text{S4_MIN}, \text{S4_MAX}\right) = \text{clip}\left(0.60 + \text{gap}/400, \frac{1}{2}, \frac{6}{7}\right). \quad (8)$$

When the 4p rating lags, more 4p is scheduled (and vice versa); the bounds are the **1:1..1:6 envelope** — 4p may dominate up to six 4p iterations per 2p iteration but never starves either format. The scheduler is **error-diffused**: an accumulator $a \mathrel{+}= s_4$ each iteration, playing 4p iff $a \geq 1$ (then $a \mathrel{-}= 1$), so the *realized* ratio equals the target exactly over any window (a Bernoulli draw would only match in expectation, with format droughts at small iteration counts).

7 League membership: invited members and weak-point matchmaking

PFSP sampling, the behavioral-fingerprint diversity term, snapshot pruning, and the scripted-anchor boosts are unchanged (see `rl_math.v6.tex § league`). 4p seats are $k=3$ weighted draws *without* replacement from the same per-format weights. Two additions:

(a) **Invited members.** The first existing folder in `INVITE_DIRS` is scanned for `*.pt`; the top `INVITE_MAX=4` by anchored rating join the league as fixed high-level bots. The anchor resolves as `filename suffix _elo<NNNN> > ckpt meta elo > INVITE_DEFAULT_ELO=750` (the submitted `it1041` leaderboard level), and never updates. While unmastered they sample at $\times \text{INVITE_MATCH_BOOST}=2$ (high-level sparring focus); once the learner’s per-format score EMA passes `INVITE_MASTER_WR=0.75` their rate *drops* to $\times \text{INVITE_MASTERED_W}=0.2$ — a small anti-forgetting floor rather than continued compute spend. Per-member architecture comes from the ckpt’s own config (smaller invited nets are fine); v7/v8-format members get the §2 obs adapter. Caveat (measured): a ckpt whose *meta* carries `elo:0` anchors at 0 — the default only applies when the meta key is absent; curate invite folders with `_elo` suffixes (the league export writes them).

(b) **Weak-point matchmaking.** Per member and per format, compare the learner’s *actual* score EMA w against the Elo-*expected* score $\mathbb{E}$. When reality runs below theory the member is a measured weak point and its sampling weight is boosted:

$$\beta(m, \text{fmt}) = \begin{cases} 1 & n < \text{WEAK_BOOST_MIN_GAMES} \text{ or } \mathbb{E} - w < \text{WEAK_BOOST_MIN_GAP} \\ \min(\text{WEAK_BOOST_MAX}, 1 + \text{WEAK_BOOST_GAIN}(\mathbb{E} - w)) & \text{otherwise,} \end{cases} \quad (9)$$

with (5, 0.05, 10, 3): e.g. Elo expects 0.90 but the EMA reads 0.70 $\Rightarrow$ gap 0.20 $\Rightarrow \times 3$ (the cap). It fires *only* on underperformance — overperforming opponents are never down-weighted — and the deadband keeps EMA noise from triggering it. Active boosts are surfaced on the leaderboard as `weak2/weak4 x<mult>`.

8 Mixed best-checkpoint gauntlet

v6 selected “best” on the fixed-opponent *2p* gauntlet (see `rl_math.v6.tex § league`); that rule is what crowned `it1041` while it was losing FFAs. v9 selects on the mix it will be graded on:

$$g = (1 - w_4) g_2 + w_4 g_4, \quad w_4 = \text{GAUNTLET_4P_W} = 0.5, \quad (10)$$

where g_2 is the v6 2p gauntlet (greedy play vs. **starter/intermediate/greedy** on held-out worlds) and g_4 is the mean *pairwise* score (Eq. (5) left) of a 4p FFA vs. the fixed trio **{starter, greedy, intermediate}** on the same held-out worlds. A 2p-only specialist can no longer win the checkpoint race.

9 Anchor scale: leaderboard-calibrated

The scripted anchors were re-pinned from the in-house round-robin scale to the *submitted* Kaggle leaderboard levels, so the learner’s rating reads directly against the competition:

anchor	old	new	basis
random	0	0	scale pin
starter	600	450	submitted starter $\approx$ 450
intermediate	750	475	never submitted; interpolated (keeps in-house ordering)
greedy	900	500	submitted greedy $\approx$ 500
medium	1000	500	submitted medium $\approx$ 500
ADVANCE_TRIGGER/CONFIRM	800/700	525/425	± 50 around intermediate , as before
INVITE_DEFAULT_ELO	1100	750	= submitted it1041 level

Two consequences to keep in mind. (i) The compressed scale *understates in-house win probabilities*: starter→medium is a $\sim 90\%$ -win gap in-house but is now encoded as 50 Elo ($\approx 57\%$), so PFSP/weak-boost *expected* scores flatten toward $\frac{1}{2}$ — the weak-point detector compensates since it compares against the same $\mathbb{E}$. (ii) League state stores member ratings: **resuming a pre-rescale train-state keeps the old anchors**; fresh runs (or warm-starts without league state) use the new ones.

10 Deployment constraint

The v7-era Kaggle adapters map *every* opponent to the single enemy owner before encoding — correct for v7/v8 (it *is* their training view) and **wrong for v9**: it would zero the seat channels’ information and discard the trained per-enemy behavior. The v9 submission adapter must *preserve seats*: map each opponent id to its own owner 1..3 (any fixed assignment — Eq. (2) is ego-relative, so only consistency matters within a game), set **n_players**= 4 in 4p episodes, and = 2 heads-up. v9 checkpoints ($F=104$) do not load in the v7-era eval tools.

11 Verification battery

- **scripts/test_v8_parity.py**: v7 vs. v8, 70 ticks (comet spawn included), identical random gated-alloc actions vs. scripted **medium** — planet/fleet state **and** encodings bit-exact (v8 is a strict superset of v7 on the 2p path).
- **scripts/test_v9_parity.py**: v8 vs. v9 — 2p *state* bit-exact; encodings bit-equal on all shared features; the one-hot exactly consistent with the scalar (self $\leftrightarrow$ 1, enemy $\leftrightarrow$ 2); channels 2/3 provably silent in 2p.
- **scripts/verify_enemy_id.py**: on a 4-seat state, v8 reads all enemies as code 2 (by design); v9 reads 0100/0010/0001 and is rotation-equivariant from every ego seat.
- **scripts/smoke_v8_nb.py** / **scripts/smoke_v9_nb.py**: the full notebooks at SMOKE scale (BC $\rightarrow$ mixed 2p/4p league $\rightarrow$ resume): both formats scheduled within the 1:1..1:6 envelope, KL inside the guardrail band, dual ratings and the mixed gauntlet logged; v9 additionally asserts $F=104$, the legacy wrapper on $F=101$ invites, and wrapper survival across the resume round-trip.

12 Config $\leftrightarrow$ symbols (delta only)

New or changed knobs (v8 config block + the v9 observation); everything else as in `rl_design_v7.tex` / `rl_math_v6.tex`.

config	symbol / role	value
N_BODY_FEATURES, F	ownership one-hot (v9 only)	14, $F=104$ (v8: 11, 101)
FOURP_ENABLED, B_4P	4p training, envs per 4p iter	on, B
OUTCOME_4P	Ω rule, Eq. (5)	placement (winner = official)
S4_BASE/MIN/MAX/GAIN	match mix, Eq. (8)	0.60 / 0.50 / 6/7 / 1.0
ELO_K4	K_4 (4p K-factor, split $/(N-1)$)	32
ELO_COUPLING	κ (cross-format coupling)	0.25
INVITE_MAX/DEFAULT_ELO/ELO4_OFFSET	invited members	4 / 750 / 0
INVITE_MATCH_BOOST/MASTER_WR/MASTERED_W	sparring focus	2.0 / 0.75 / 0.2
WEAK_BOOST_MIN_GAMES/MIN_GAP/GAIN/MAX	weak-point boost β	5 / 0.05 / 10 / 3
GAUNTLET_4P_W	w_4 (mixed best-ckpt weight)	0.5
ELO_STARTER/INTERMEDIATE/GREEDY/MEDIUM	anchor ratings	450 / 475 / 500 / 500
ADVANCE_TRIGGER_ELO/CONFIRM_ELO	curriculum gates	525 / 425